

Human Body Systems: Digestion & Control

Answer sheet · 37 marks total

Section A — Digestion & enzymes

A1. Liver (1).

A2. Biological catalyst / protein (1).

A3. Amylase → sugars (2); protease → amino acids (2); lipase → fatty acids + glycerol (2).

A4. Any two, 2 marks each: neutralises stomach acid (1), giving the right/alkaline pH for enzymes in the small intestine (1); emulsifies fats into tiny droplets (1), increasing the surface area for lipase to act on (1).

Section B — The nervous system

B1. Sensory neuron (1), relay neuron (1), motor neuron (1).

B2. The gap between two neurons (1); a chemical is released to carry the signal across it, triggering a new impulse in the next neuron (1).

B3. They produce a very fast response (1), protecting the body from harm/damage before the brain has time to process the stimulus (1).

B4. The impulse is processed by a relay neuron in the spinal cord (1) rather than travelling to the brain and back, so no conscious thought/decision-making is needed (1).

Section C — The endocrine system

C1. A chemical messenger (1), released directly into the blood by a gland, that produces an effect at a target organ (1).

C2. Pituitary gland (1).

C3. Insulin (1); produced by the pancreas (1).

C4. Any two, 2 marks each: nervous signals are electrical, hormonal signals are chemical (1) [explanation (1)]; nervous is fast-acting, hormonal is slower (1) [explanation (1)]; nervous effects are short-lived, hormonal effects last longer (1) [explanation (1)].

Section D — Apply it

D1. The stimulus (heat/pain) is detected by a receptor and passed along a sensory neuron to the spinal cord (1); a relay neuron passes the impulse straight to a motor neuron, without going via the brain (1); this means the muscle (effector) responds before the brain registers pain, because it is a reflex, not a conscious decision (1).

D2. The pancreas normally produces insulin (1); insulin reduces blood glucose concentration, e.g. by causing glucose to be stored as glycogen (1); a damaged pancreas cannot produce enough insulin, so glucose is not removed from the blood effectively and levels stay high (1).

Marking note: accept any scientifically correct alternative answer with the same reasoning, even if different from the example shown — award marks for a correctly linked point, not for matching these exact words.